

A Local Two-Dimensional $\mathbb{Z}_2$ Marker for a Spin-Mixed BHZ-Type Model

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SUMMARY AND ROADMAP

The purpose of this note is to identify a local time-reversal-odd operator S for the two-dimensional marker, and to understand how the appropriate choice of S changes as increasingly general forms of spin mixing are introduced. The main results can be summarized as follows.

Sections 2–9 consider the original two-dimensional model. Although the physical spin s_z is not conserved, the orbital-dressed operator

$$\boxed{S_0 = \tau_z s_z} \quad (1)$$

is exactly conserved. It is local, Hermitian, traceless, time-reversal odd, and satisfies $S_0^2 = 1$. Consequently the corresponding auxiliary operator $\tilde{S} = \{\rho, S_0\} - S_0$ has the maximal possible spectral gap. The construction can be understood geometrically: in two dimensions all spin-orbit hopping vectors lie in the xy plane, and the s_z axis provides a common perpendicular direction once it is dressed by τ_z .

Sections 10–15 address local spin-frame disorder. When the Hamiltonian is changed only by an onsite unitary transformation,

$$H \longrightarrow H' = U H U^\dagger, \quad (2)$$

the appropriate grading is not obtained by keeping a fixed physical spin axis. Instead, S must be transformed by the same local rotation,

$$\boxed{S_0 \longrightarrow S' = U S_0 U^\dagger.} \quad (3)$$

The transformed operator remains local and exactly conserved, and the spectrum of $\tilde{S}$ is unchanged. Thus local spin-frame scrambling of this type does not destroy the grading; it merely rotates its local representation. The local marker is correspondingly basis covariant.

Sections 16–21 turn to genuine spin mixing, for which the Hamiltonian is no longer related to the original model by a local basis rotation. Rashba spin-orbit coupling provides the main example. In this case S_0 is no longer conserved, so the relevant criterion is not $[H, S] = 0$ but rather the size of the spectral gap of $\tilde{S}$. The strategy is to solve the clean, translation-invariant problem first, where the optimization reduces to 4×4 matrices at each momentum. For Rashba coupling this gives the one-parameter family

$$S(\theta) = \cos \theta \tau_z s_z + \sin \theta \tau_y \quad (4)$$

and the exact clean optimum

$$\boxed{\theta_{\text{opt}}^{\text{clean}} = \frac{1}{2} \arctan \frac{\lambda_R}{A}.} \quad (5)$$

This construction has a natural extension to an amorphous lattice. The operator $S(\theta)$ is strictly onsite and therefore remains well defined when momentum and translation symmetry are lost. More importantly, the algebra that selects the two matrices $\tau_z s_z$ and τ_y survives arbitrary bond orientations. For a bond at angle ϕ_{ij} , the original BHZ spin-orbit hopping contains the spin component $s_{\parallel}(\phi_{ij})$ along the bond, while the Rashba term contains the perpendicular component $s_{\perp}(\phi_{ij})$. Their commutators obey, bond by bond,

$$[S_0, T_{ij}^R] = -\frac{\lambda_R}{A} [\tau_y, T_{ij}^{\text{SO}}], \quad (6)$$

independently of the bond angle and bond length, provided the two hopping channels have the same radial envelope and a fixed ratio λ_R/A . Structural disorder therefore changes the geometry of the bonds without introducing a new internal matrix direction into the grading problem. This is why the same local family $S(\theta)$ remains the natural ansatz on an amorphous lattice.

What is lost in the amorphous system is the momentum-space argument that fixes the exact midpoint angle. The clean result therefore provides a physically motivated starting point, while the actual amorphous optimum is determined directly in real space by constructing the occupied-state projector ρ_{amor} and maximizing

$$\Delta_{\tilde{S}}^{\text{amor}}(\theta) = \min_a |\lambda_a [\{\rho_{\text{amor}}, S(\theta)\} - S(\theta)]|. \quad (7)$$

Thus the crystalline calculation identifies the relevant local degrees of freedom in S , while the amorphous calculation determines how the optimal balance between them is modified by a particular structural-disorder realization.

Finally, Sec. 22 adds Dresselhaus spin-orbit coupling. Whereas the Rashba texture has the continuum form $k_y s_x - k_x s_y$, the linear Dresselhaus texture has the form $k_x s_x - k_y s_y$. Combining the two destroys the pointwise orthogonality that makes the pure Rashba problem especially simple, and the simple Rashba conserved sector operator no longer exists. Nevertheless, the flattened Hamiltonian retains enough algebraic structure that the auxiliary spectrum can again be written as

$$\delta(\mathbf{k}, \theta) = |\cos[\theta - \vartheta(\mathbf{k})]|. \quad (8)$$

The range of the momentum-dependent preferred angle $\vartheta(\mathbf{k})$ can be obtained analytically, giving both the optimal fixed onsite grading

$$\theta_{\text{opt}} = \frac{\vartheta_{\min} + \vartheta_{\max}}{2} \quad (9)$$

and its optimal auxiliary gap

$$\Delta_{\tilde{S}}^{\text{opt}} = \cos \left[\frac{\vartheta_{\max} - \vartheta_{\min}}{2} \right]. \quad (10)$$

Thus the same basic strategy extends from an exactly conserved grading, through Rashba spin mixing and structural disorder, to a more general Rashba–Dresselhaus spin texture.

I. GOAL

Ref: [1] claim: *Specifically, the spin-Bott and Chern markers fail because locally scrambling the spin generically closes the spin gap in all directions. In some realistic crystalline scenarios, spin-scrambling terms like Rashba spin-orbit coupling can be small enough to keep the gap open [94], but it is unlikely that in amorphous systems spin-frame disorder remains small. Moreover, the spin-gap is always closed in three-dimensional Z2 topological insulators [116]. Because the existence of a global spin gap is required for these operators to be well-defined, they thus fail to diagnose topology.*

We want to implement in two dimensions the local time-reversal-invariant marker

$$\nu_{\text{TRI}}(\mathbf{r}) = \frac{\pi}{8i} \sum_{\alpha} \sum_{i,j=x,y} \epsilon_{ij} \left[\frac{\tilde{S}}{|\tilde{S}|} X_i \frac{\tilde{S}}{|\tilde{S}|} X_j \frac{\tilde{S}}{|\tilde{S}|} \right]_{(\mathbf{r},\alpha),(\mathbf{r},\alpha)} \mod 2, \quad (11)$$

where

$$\tilde{S} = \{\rho, S\} - S. \quad (12)$$

(I use the opposite sign from the chiral marker that we, this produces the correct spin Chern marker... in the derivation. It is just a convention, but I believe after many checks that we defined the chiral markers with opposite sign to the chiral winding number in Ryu etc.) Here ρ is the occupied-state projector in the noninteracting problem. More generally, in an interacting formulation one may replace it by the appropriately band-flattened one-particle density matrix.

The operator S must satisfy the following conditions:

$$S^{\dagger} = S, \quad (13)$$

$$S^2 = 1, \quad (14)$$

$$\text{Tr } S = 0, \quad (15)$$

$$TST^{-1} = -S, \quad (16)$$

and, crucially, $\tilde{S}$ must have no zero eigenvalues,

$$\text{gap}(\tilde{S}) \equiv \min_a |\lambda_a(\tilde{S})| > 0. \quad (17)$$

The main practical question is therefore:

Can one identify a simple local operator S satisfying Eqs. (13)–(16) for a model with substantial spin mixing, while simultaneously keeping the spectrum of $\tilde{S}$ gapped?

The two-dimensional model of Ref. [1] provides a particularly useful test case.

II. THE TWO-DIMENSIONAL MODEL OF REF. [1]

A. Local Hilbert space

The model is defined on a set of real-space positions $\{\mathbf{r}_j\}$. Each site contains two orbital degrees of freedom and a spin-1/2 degree of freedom, giving four internal states per site. We denote the Pauli matrices acting on orbital space by τ_j , and the Pauli matrices acting on physical spin by s_j where $j = (x, y, z)$, where operators acting on these two spaces commute, $[\tau_\mu, s_\nu] = 0$. We suppress explicit tensor-product symbols and write $\tau_x s_y \equiv \tau_x \otimes s_y$.

B. Hamiltonian

The Hamiltonian considered in Ref. [1] is

$$H = \sum_{\langle jk \rangle} f_{jk} |\mathbf{r}_j\rangle \langle \mathbf{r}_k| \otimes t_{jk} + \sum_j |\mathbf{r}_j\rangle \langle \mathbf{r}_j| \otimes \epsilon_j. \quad (18)$$

f_{jk} controls the hopping strength, and depends on the distance between sites:

$$f_{jk} = \exp \left[-\frac{\delta_{jk} - r_0}{r_0} \right] \Theta(R - \delta_{jk}), \quad (19)$$

where

$$\delta_{jk} = \mathbf{r}_j - \mathbf{r}_k, \quad \delta_{jk} = |\delta_{jk}|, \quad \hat{\delta}_{jk} = \frac{\delta_{jk}}{\delta_{jk}}. \quad (20)$$

The internal hopping matrix is

$$t_{jk} = \frac{A}{2} \left[\tau_z s_0 - i\tau_x \left(\hat{\delta}_{jk} \cdot \mathbf{s} \right) \right]. \quad (21)$$

In two dimensions,

$$\hat{\delta}_{jk} \cdot \mathbf{s} = \hat{\delta}_{jk,x} s_x + \hat{\delta}_{jk,y} s_y. \quad (22)$$

The on-site energy is

$$\epsilon_j = M\tau_z s_0 + W_j\tau_0 s_0, \quad (23)$$

where M is an orbital asymmetry and the random scalar W_j describes on-site Anderson disorder, drawn from a uniform distribution $W_j \in [-w/2, W/2]$.

The important feature for our purposes is the term

$$\tau_x \left(\hat{\delta}_{jk,x} s_x + \hat{\delta}_{jk,y} s_y \right), \quad (24)$$

which explicitly mixes physical spin components.

III. WHY THE PHYSICAL SPIN s_z IS NOT CONSERVED

Consider first the conventional choice

$$S_{\text{spin}} = s_z. \quad (25)$$

The spin-independent hopping term commutes with s_z , $[s_z, \tau_z] = 0$. However, $[s_z, \tau_x s_x] = \tau_x [s_z, s_x] = 2i\tau_x s_y$, and $[s_z, \tau_x s_y] = \tau_x [s_z, s_y] = -2i\tau_x s_x$.

Therefore $[s_z, H] \neq 0$ for the generic two-dimensional Hamiltonian Eq. (18)

This is the reason why a conventional spin-Chern or spin-Bott construction based on a globally fixed physical spin axis is potentially problematic. Ref. [1] discusses this problem in terms of the spectral gap of the projected spin operator $\rho s_z \rho$. If this projected-spin gap closes, its positive- and negative-eigenvalue sectors can no longer be used to define a stable decomposition into two effective Chern sectors. Ref. [1] therefore employs a spectral-localizer construction that does not require selecting a conserved physical spin component.

IV. AN EXACT LOCAL OPERATOR FOR THE TWO-DIMENSIONAL MODEL

We now turn to the local marker in Eq. (11).

The central observation is that, although s_z itself is not conserved, the orbital-dressed spin operator

$$\boxed{S_0 = \tau_z s_z} \quad (26)$$

is exactly conserved by the two-dimensional Hamiltonian (18).

We verify all required properties explicitly.

A. Hermiticity

Since both τ_z and s_z are Hermitian,

$$S_0^\dagger = (\tau_z s_z)^\dagger = \tau_z s_z = S_0. \quad (27)$$

B. S_0 squares to unity

Because orbital and spin operators commute,

$$\begin{aligned} S_0^2 &= (\tau_z s_z)(\tau_z s_z) \\ &= \tau_z^2 s_z^2 \\ &= 1. \end{aligned} \quad (28)$$

C. Vanishing trace

The internal trace factorizes:

$$\begin{aligned} \text{Tr}_{\text{int}} S_0 &= \text{Tr}_\tau(\tau_z) \text{Tr}_s(s_z) \\ &= 0. \end{aligned} \quad (29)$$

D. Time-reversal transformation

For the model in its conventional basis, time reversal acts as

$$T = i s_y K. \quad (30)$$

The orbital operator τ_z is time-reversal even,

$$T\tau_zT^{-1} = \tau_z, \quad (31)$$

whereas physical spin is time-reversal odd,

$$Ts_zT^{-1} = -s_z. \quad (32)$$

Therefore

$$\begin{aligned} TS_0T^{-1} &= T(\tau_zs_z)T^{-1} \\ &= \tau_z(-s_z) \\ &= -S_0. \end{aligned} \quad (33)$$

Thus S_0 has precisely the time-reversal transformation required by the marker.

V. EXACT COMMUTATION WITH THE TWO-DIMENSIONAL HAMILTONIAN

The important result is

$$\boxed{[S_0, H] = 0.} \quad (34)$$

We now show this term by term.

A. On-site terms

The mass term gives

$$[\tau_zs_z, \tau_z] = 0. \quad (35)$$

The scalar disorder term trivially gives

$$[\tau_zs_z, \tau_0s_0] = 0. \quad (36)$$

Hence

$$[S_0, \epsilon_j] = 0 \quad (37)$$

for every site j , independently of M and W_j .

B. Spin-independent hopping

The first part of the hopping matrix is proportional to τ_zs_0 . Therefore

$$[S_0, \tau_zs_0] = 0. \quad (38)$$

C. Spin-mixing hopping

The nontrivial part is

$$\tau_xs_a, \quad a = x, y. \quad (39)$$

Both the orbital and spin factors anticommute:

$$\{\tau_z, \tau_x\} = 0, \quad (40)$$

and

$$\{s_z, s_x\} = \{s_z, s_y\} = 0. \quad (41)$$

Consider first $\tau_x s_x$. We have

$$(\tau_z s_z)(\tau_x s_x) = \tau_z \tau_x s_z s_x. \quad (42)$$

In the reverse order,

$$\begin{aligned} (\tau_x s_x)(\tau_z s_z) &= \tau_x \tau_z s_x s_z \\ &= (-\tau_z \tau_x)(-s_z s_x) \\ &= \tau_z \tau_x s_z s_x. \end{aligned} \quad (43)$$

Comparing Eqs. (42) and (43),

$$[\tau_z s_z, \tau_x s_x] = 0. \quad (44)$$

Exactly the same argument gives

$$[\tau_z s_z, \tau_x s_y] = 0. \quad (45)$$

Therefore

$$\left[\tau_z s_z, \tau_x \left(\hat{\delta}_{jk,x} s_x + \hat{\delta}_{jk,y} s_y \right) \right] = 0 \quad (46)$$

for every bond direction $\hat{\delta}_{jk}$.

Hence

$$[S_0, t_{jk}] = 0 \quad (47)$$

for every bond jk .

D. Consequences for structural disorder

The scalar coefficient f_{jk} multiplying the hopping matrix does not act on the internal Hilbert space. Consequently, changing the bond length changes f_{jk} but cannot affect the commutation relation,

$$[S_0, f_{jk} t_{jk}] = f_{jk} [S_0, t_{jk}] = 0. \quad (48)$$

Similarly, structural disorder changes the positions $\mathbf{r}_j$, the set of connected bonds, and the bond directions $\hat{\delta}_{jk}$.

However, in two dimensions every bond direction still has only x and y components. Since S_0 commutes separately with both $\tau_x s_x$ and $\tau_x s_y$, arbitrary structural disorder of the form considered in Ref. [1] preserves Eq. (34).

Thus the exact symmetry survives:

- arbitrary two-dimensional bond directions;
- arbitrary connectivity;
- bond-length-dependent hopping amplitudes f_{jk} ;
- scalar on-site Anderson disorder W_j .

VI. THE $\tilde{S}$ GAP IS EXACTLY MAXIMAL

Let ρ denote the spectral projector onto the occupied single-particle states,

$$\rho = \Theta(E_F - H). \quad (49)$$

Since S_0 commutes with H , it also commutes with any spectral function of H . Therefore

$$[S_0, \rho] = 0. \quad (50)$$

Using Eq. (12),

$$\begin{aligned} \tilde{S} &= \{\rho, S_0\} - S_0 \\ &= \rho S_0 + S_0 \rho - S_0. \end{aligned} \quad (51)$$

Since $[\rho, S_0] = 0$,

$$\begin{aligned} \tilde{S} &= 2S_0\rho - S_0 \\ &= S_0(2\rho - 1). \end{aligned} \quad (52)$$

Now

$$S_0^2 = 1, \quad (53)$$

and, because ρ is a projector,

$$\rho^2 = \rho \implies (2\rho - 1)^2 = 1. \quad (54)$$

Furthermore,

$$[S_0, 2\rho - 1] = 0. \quad (55)$$

Consequently,

$$\begin{aligned} \tilde{S}^2 &= S_0^2(2\rho - 1)^2 \\ &= 1. \end{aligned} \quad (56)$$

Therefore

$$\boxed{\text{spec}(\tilde{S}) = \{+1, -1\}.} \quad (57)$$

In particular,

$$\boxed{\min_a |\lambda_a(\tilde{S})| = 1.} \quad (58)$$

Thus the spectral gap of $\tilde{S}$ around zero is not merely finite: it is maximal.

VII. SIMPLIFICATION OF THE LOCAL MARKER

Since

$$\tilde{S}^2 = 1, \quad (59)$$

we have

$$|\tilde{S}| = 1, \quad (60)$$

and therefore

$$\frac{\tilde{S}}{|\tilde{S}|} = \tilde{S} = S_0(2\rho - 1). \quad (61)$$

The general marker in Eq. (11) then reduces to the commuting- S form

$$\nu_{\text{TRI}}(\mathbf{r}) = \frac{1}{\pi 1} \sum_{\alpha} \sum_{i,j=x,y} \epsilon_{ij} [\rho S_0 X_i \rho S_0 X_j \rho S_0]_{(\mathbf{r},\alpha),(\mathbf{r},\alpha)} \mod 2. \quad (62)$$

This is the natural generalization of the familiar spin-Chern construction, but the role of spin is played by S_0 rather than by a physical spin component.

VIII. DECOMPOSITION INTO TWO GENERALIZED CHERN SECTORS

Because

$$[\rho, S_0] = 0, \quad (63)$$

we can simultaneously resolve the occupied space according to the eigenvalues $S_0 = \pm 1$.

Define

$$\Pi_{\pm} = \frac{1 \pm S_0}{2}. \quad (64)$$

The occupied projectors in the two sectors are

$$\rho_{\pm} = \rho \Pi_{\pm} = \rho \frac{1 \pm S_0}{2}. \quad (65)$$

These are not physical spin-up and spin-down sectors. Instead, they correspond to

$$\tau_z s_z = +1 \quad (66)$$

and

$$\tau_z s_z = -1. \quad (67)$$

Because time reversal leaves ρ invariant,

$$T \rho T^{-1} = \rho, \quad (68)$$

but reverses S_0 ,

$$T S_0 T^{-1} = -S_0, \quad (69)$$

we find

$$\begin{aligned} T \rho_{\pm} T^{-1} &= T \rho \frac{1 \pm S_0}{2} T^{-1} \\ &= \rho \frac{1 \mp S_0}{2} \\ &= \rho_{\mp}. \end{aligned} \quad (70)$$

Thus time reversal exchanges the two generalized Chern sectors.

Their Chern numbers therefore satisfy

$$C_- = -C_+. \quad (71)$$

The $\mathbb{Z}_2$ index can equivalently be viewed as

$$\nu = \frac{1}{2} (C_+ - C_-) \bmod 2. \quad (72)$$

Since $C_- = -C_+$,

$$\nu = C_+ \bmod 2. \quad (73)$$

The important difference from the conventional spin-Chern construction is that the decomposition is generated by the conserved dressed spin

$$S_0 = \tau_z s_z \quad (74)$$

rather than by s_z .

IX. A MORE GENERAL GEOMETRICAL ANSATZ

The structure of the exact solution suggests a slightly more general class of trial operators. Consider

$$S(\hat{\mathbf{n}}) = \tau_z (\hat{\mathbf{n}} \cdot \mathbf{s}), \quad (75)$$

where

$$|\hat{\mathbf{n}}| = 1. \quad (76)$$

This automatically satisfies

$$S(\hat{\mathbf{n}})^2 = 1 \quad (77)$$

and

$$TS(\hat{\mathbf{n}})T^{-1} = -S(\hat{\mathbf{n}}). \quad (78)$$

Now consider a generic hopping term

$$\tau_x(\mathbf{d} \cdot \mathbf{s}). \quad (79)$$

Its commutator with Eq. (75) is

$$\begin{aligned} & [\tau_z(\hat{\mathbf{n}} \cdot \mathbf{s}), \tau_x(\mathbf{d} \cdot \mathbf{s})] \\ &= \tau_z \tau_x (\hat{\mathbf{n}} \cdot \mathbf{s})(\mathbf{d} \cdot \mathbf{s}) - \tau_x \tau_z (\mathbf{d} \cdot \mathbf{s})(\hat{\mathbf{n}} \cdot \mathbf{s}) \\ &= \tau_z \tau_x [(\hat{\mathbf{n}} \cdot \mathbf{s})(\mathbf{d} \cdot \mathbf{s}) + (\mathbf{d} \cdot \mathbf{s})(\hat{\mathbf{n}} \cdot \mathbf{s})] \\ &= \tau_z \tau_x \{\hat{\mathbf{n}} \cdot \mathbf{s}, \mathbf{d} \cdot \mathbf{s}\}. \end{aligned} \quad (80)$$

Using

$$\{s_i, s_j\} = 2\delta_{ij}, \quad (81)$$

we obtain

$$\boxed{[\tau_z(\hat{\mathbf{n}} \cdot \mathbf{s}), \tau_x(\mathbf{d} \cdot \mathbf{s})] = 2(\hat{\mathbf{n}} \cdot \mathbf{d})\tau_z\tau_x.} \quad (82)$$

Therefore a sufficient condition for conservation is

$$\boxed{\hat{\mathbf{n}} \cdot \mathbf{d}_{jk} = 0 \quad \text{for every bond } jk.} \quad (83)$$

For the two-dimensional model,

$$\mathbf{d}_{jk} = (\hat{\delta}_{jk,x}, \hat{\delta}_{jk,y}, 0). \quad (84)$$

All spin-orbit vectors therefore lie in the xy plane. A common perpendicular direction exists:

$$\hat{\mathbf{n}} = \hat{\mathbf{z}}. \quad (85)$$

Equation (75) then becomes

$$S(\hat{\mathbf{z}}) = \tau_z s_z, \quad (86)$$

which reproduces the exact solution.

This also explains why the construction is naturally two-dimensional.

In the three-dimensional version of the model of Ref. [1], the hopping contains all three physical spin components,

$$s_x, \quad s_y, \quad s_z. \quad (87)$$

Generically there is then no nonzero vector $\hat{\mathbf{n}}$ orthogonal to every spin-orbit vector $\mathbf{d}_{jk}$. The simple ansatz (75) therefore need not produce an exact conserved operator in three dimensions.

X. LOCAL SPIN-FRAME ROTATIONS

Ref. [1] additionally studies local spin-frame disorder.

The important point is that this disorder is implemented as an on-site unitary transformation,

$$U = \sum_r |\mathbf{r}\rangle\langle\mathbf{r}| \otimes U_r. \quad (88)$$

The Hamiltonian is transformed as

$$H' = U H U^\dagger. \quad (89)$$

Since U acts locally in internal space,

$$[U, X] = [U, Y] = 0. \quad (90)$$

A. Transformation of time reversal

If initially

$$T = wK, \quad (91)$$

then after the basis transformation,

$$T' = U T U^\dagger. \quad (92)$$

Using the antiunitarity of T , this can be written as

$$T' = w'K, \quad (93)$$

with

$$w' = U w U^T. \quad (94)$$

This is the basis-covariance emphasized in Ref. [1].

XI. THE APPROPRIATE S TRANSFORMS WITH THE LOCAL FRAME

Rather than keeping S_0 fixed after the local basis change, we should transform it covariantly:

$$\boxed{S' = U S_0 U^\dagger}. \quad (95)$$

We now verify that all desired properties survive.

A. Locality

Since U is an on-site direct sum and S_0 is on-site,

$$S' = \sum_r |\mathbf{r}\rangle\langle\mathbf{r}| \otimes S'_r, \quad (96)$$

where

$$S'_r = U_r S_0 U_r^\dagger. \quad (97)$$

Thus S' remains strictly local.

B. Square

We have

$$\begin{aligned}
 (S')^2 &= US_0U^\dagger US_0U^\dagger \\
 &= US_0^2U^\dagger \\
 &= 1.
 \end{aligned} \tag{98}$$

C. Time-reversal oddness

Using

$$T' = UTU^\dagger, \tag{99}$$

we obtain

$$\begin{aligned}
 T'S'T'^{-1} &= (UTU^\dagger)(US_0U^\dagger)(UT^{-1}U^\dagger) \\
 &= U(TS_0T^{-1})U^\dagger \\
 &= -US_0U^\dagger \\
 &= -S'.
 \end{aligned} \tag{100}$$

D. Commutation with the transformed Hamiltonian

Similarly,

$$\begin{aligned}
 [S', H'] &= [US_0U^\dagger, UHU^\dagger] \\
 &= U[S_0, H]U^\dagger \\
 &= 0.
 \end{aligned} \tag{101}$$

Thus an arbitrary local basis scrambling cannot destroy the existence of the exact conserved S ; it merely changes its local representation.

XII. EXPLICIT HADAMARD SPIN-FRAME DISORDER

Ref. [1] uses a Hadamard transformation as a simple example of local spin-frame disorder. To avoid confusing the Hadamard matrix with the Hamiltonian, we denote it by $\mathcal{H}$:

$$\mathcal{H} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \tag{102}$$

At each site one chooses

$$a_r \in \{0, 1\}, \tag{103}$$

and defines

$$U_r = \mathcal{H}^{a_r}. \tag{104}$$

Thus

$$U = \sum_r |\mathbf{r}\rangle\langle\mathbf{r}| \otimes \mathcal{H}^{a_r}. \tag{105}$$

The Hadamard matrix satisfies

$$\mathcal{H}s_z\mathcal{H}^\dagger = s_x. \tag{106}$$

Since it acts only in spin space,

$$[\mathcal{H}, \tau_z] = 0. \quad (107)$$

The transformed S therefore has the explicit local form

$$S' = \sum_r |\mathbf{r}\rangle \langle \mathbf{r}| \otimes \tau_z \begin{cases} s_z, & a_r = 0, \\ s_x, & a_r = 1. \end{cases} \quad (108)$$

Thus the appropriate local pseudospin axis can vary from site to site even though the operator remains a perfectly valid local S .

The physical-spin operator s_z may then appear completely scrambled, while the dressed and covariantly transformed operator S' remains exactly conserved.

XIII. THE $\tilde{S}$ GAP UNDER FRAME DISORDER

The occupied-state projector transforms as

$$\rho' = U \rho U^\dagger. \quad (109)$$

The corresponding $\tilde{S}$ operator is

$$\begin{aligned} \tilde{S}' &= \{\rho', S'\} - S' \\ &= \{U \rho U^\dagger, U S_0 U^\dagger\} - U S_0 U^\dagger \\ &= U (\{\rho, S_0\} - S_0) U^\dagger \\ &= U \tilde{S} U^\dagger. \end{aligned} \quad (110)$$

Therefore $\tilde{S}'$ has exactly the same spectrum as $\tilde{S}$.
Since

$$\tilde{S}^2 = 1, \quad (111)$$

we also have

$$(\tilde{S}')^2 = 1. \quad (112)$$

Hence

$$\boxed{\min_a |\lambda_a(\tilde{S}')| = 1} \quad (113)$$

for any spin-frame realization that is generated purely by such local basis rotations.

XIV. BASIS COVARIANCE OF THE LOCAL MARKER

Define

$$\mathcal{A} = \frac{\tilde{S}}{|\tilde{S}|}. \quad (114)$$

Under the local transformation,

$$\mathcal{A}' = U \mathcal{A} U^\dagger. \quad (115)$$

Since

$$[U, X_i] = 0, \quad (116)$$

we find

$$\begin{aligned}\mathcal{A}'X_i\mathcal{A}'X_j\mathcal{A}' &= U\mathcal{A}U^\dagger X_i U\mathcal{A}U^\dagger X_j U\mathcal{A}U^\dagger \\ &= U\mathcal{A}X_i\mathcal{A}X_j\mathcal{A}U^\dagger.\end{aligned}\tag{117}$$

The local marker requires a trace over the internal degrees of freedom at fixed real-space position $\mathbf{r}$. Denoting this local internal trace by $\text{Tr}_{\text{int},\mathbf{r}}$,

$$\begin{aligned}\text{Tr}_{\text{int},\mathbf{r}} [U\mathcal{A}X_i\mathcal{A}X_j\mathcal{A}U^\dagger] \\ &= \text{Tr}_{\text{int},\mathbf{r}} [U_r (\mathcal{A}X_i\mathcal{A}X_j\mathcal{A})_{\mathbf{r}\mathbf{r}} U_r^\dagger] \\ &= \text{Tr}_{\text{int},\mathbf{r}} [(\mathcal{A}X_i\mathcal{A}X_j\mathcal{A})_{\mathbf{r}\mathbf{r}}].\end{aligned}\tag{118}$$

Therefore

$$\boxed{\nu'_{\text{TRI}}(\mathbf{r}) = \nu_{\text{TRI}}(\mathbf{r})}.\tag{119}$$

Thus the marker is pointwise invariant under arbitrary on-site changes of internal basis, provided S and the representation of time reversal are transformed consistently.

XV. RELATION TO THE FAILURE OF THE CONVENTIONAL SPIN-BOTT INDEX

The distinction is now transparent.

A conventional spin-Bott construction chooses one fixed physical spin operator, for example

$$s_z,\tag{120}$$

and constructs the projected-spin operator

$$\rho s_z \rho.\tag{121}$$

Under strong local spin-frame scrambling, the fixed operator s_z no longer tracks the local internal frame. Its projected-spin gap can therefore close.

This is the mechanism demonstrated numerically in Ref. [1].

Our construction is different.

We do not require S to be a fixed physical spin component. Instead, it is sufficient that S be

1. local,
2. Hermitian,
3. an involution, $S^2 = 1$,
4. time-reversal odd,
5. and such that $\tilde{S} = \{\rho, S\} - S$ remains gapped.

For the present model the natural operator is

$$S_0 = \tau_z s_z.\tag{122}$$

Under local frame rotations it changes to

$$S' = U S_0 U^\dagger.\tag{123}$$

The construction therefore follows the local basis rather than insisting on a globally fixed physical spin direction.

XVI. WHAT HAPPENS FOR GENUINE SYMMETRY-BREAKING SPIN MIXING?

The frame disorder above is a basis transformation:

$$H \rightarrow U H U^\dagger. \quad (124)$$

A more demanding test is to add a genuinely new physical perturbation,

$$H \rightarrow H + \lambda V, \quad (125)$$

for which

$$[V, S_0] \neq 0. \quad (126)$$

Then

$$[\rho, S_0] \neq 0 \quad (127)$$

in general.

Importantly, the marker does not require exact commutation. It only requires the $\tilde{S}$ gap to remain open. For a projector ρ and an involution S , one has

$$\boxed{\tilde{S}^2 = 1 + [S, \rho]^2}. \quad (128)$$

Because S and ρ are Hermitian,

$$[S, \rho]^\dagger = -[S, \rho], \quad (129)$$

so $[S, \rho]^2$ is negative semidefinite.

Consequently the eigenvalues of $\tilde{S}^2$ lie between zero and one. Exact commutation gives

$$[S, \rho] = 0 \implies \tilde{S}^2 = 1. \quad (130)$$

Weak violation of the symmetry decreases the $\tilde{S}$ gap continuously. The marker remains well defined until an eigenvalue of $\tilde{S}$ reaches zero.

XVII. GENERAL TRANSLATION-INVARIANT ANSATZ FOR S

If a perturbation destroys exact conservation of $S_0 = \tau_z s_z$, we can enlarge the ansatz.

For a four-dimensional on-site Hilbert space with

$$T = i s_y K, \quad (131)$$

a general Hermitian time-reversal-odd on-site operator can be written as

$$S = \mathbf{n}_0 \cdot \mathbf{s} + (\mathbf{n}_1 \cdot \mathbf{s}) \tau_x + (\mathbf{n}_3 \cdot \mathbf{s}) \tau_z - m \tau_y, \quad (132)$$

where

$$\mathbf{n}_0, \mathbf{n}_1, \mathbf{n}_3 \in \mathbb{R}^3, \quad m \in \mathbb{R}. \quad (133)$$

The time-reversal transformation follows because

$$T \mathbf{s} T^{-1} = -\mathbf{s}, \quad (134)$$

while, in this basis,

$$T \tau_x T^{-1} = \tau_x, \quad T \tau_z T^{-1} = \tau_z, \quad T \tau_y T^{-1} = -\tau_y. \quad (135)$$

Thus every term in Eq. (132) is time-reversal odd.

A. Constraint $S^2 = 1$

Squaring Eq. (132) gives

$$\begin{aligned}
 S^2 = & (|\mathbf{n}_0|^2 + |\mathbf{n}_1|^2 + |\mathbf{n}_3|^2 + m^2) 1 \\
 & + 2(\mathbf{n}_0 \cdot \mathbf{n}_1)\tau_x + 2(\mathbf{n}_0 \cdot \mathbf{n}_3)\tau_z \\
 & + 2(\mathbf{n}_1 \times \mathbf{n}_3 - m\mathbf{n}_0) \cdot \mathbf{s} \tau_y.
 \end{aligned} \tag{136}$$

Therefore $S^2 = 1$ is equivalent to the constraints

$$|\mathbf{n}_0|^2 + |\mathbf{n}_1|^2 + |\mathbf{n}_3|^2 + m^2 = 1, \tag{137}$$

$$\mathbf{n}_0 \cdot \mathbf{n}_1 = 0, \tag{138}$$

$$\mathbf{n}_0 \cdot \mathbf{n}_3 = 0, \tag{139}$$

$$\mathbf{n}_1 \times \mathbf{n}_3 = m\mathbf{n}_0. \tag{140}$$

The exact solution found above corresponds simply to

$$\mathbf{n}_3 = \hat{\mathbf{z}}, \quad \mathbf{n}_0 = \mathbf{n}_1 = 0, \quad m = 0, \tag{141}$$

which gives

$$S = \tau_z s_z. \tag{142}$$

XVIII. GENUINE SPIN MIXING AND THE CHOICE OF THE LOCAL OPERATOR S

The exact operator

$$S_0 = \tau_z s_z \tag{143}$$

provides a useful starting point for the two-dimensional model discussed above. However, because $[S_0, H] = 0$ for the unperturbed model, this example does not yet probe the genuinely nonconserved- S regime for which the general construction

$$\tilde{S} = \{\rho, S\} - S \tag{144}$$

is required.

We therefore add a physical spin-mixing perturbation of Rashba type. The purpose of the following calculation is twofold. First, it gives a controlled example in which no exact S is conserved. Second, it illustrates a strategy for choosing S : rather than attempting to make S commute with the Hamiltonian, we choose S so as to maximize the spectral gap of $\tilde{S}$.

An important point is that the marker itself is a real-space construction and does not require translation invariance. We first solve the clean translation-invariant problem in momentum space only because this reduces the problem to 4×4 matrices and allows the optimal S to be obtained analytically. The resulting S is a local on-site operator and therefore remains meaningful after translation invariance is broken. We return explicitly to the real-space formulation below.

A. Real-space Hamiltonian and its Bloch representation

Let $c_{\mathbf{r}}$ denote the four-component fermion operator at site $\mathbf{r}$, with the internal Hilbert space given by orbital and spin degrees of freedom. On the clean square lattice we consider

$$\begin{aligned}
 H = & \sum_{\mathbf{r}} c_{\mathbf{r}}^\dagger M \tau_z c_{\mathbf{r}} \\
 & + \sum_{\mathbf{r}} [c_{\mathbf{r}}^\dagger t_x c_{\mathbf{r}+\hat{\mathbf{x}}} + c_{\mathbf{r}}^\dagger t_y c_{\mathbf{r}+\hat{\mathbf{y}}} + \text{h.c.}] ,
 \end{aligned} \tag{145}$$

where

$$t_x = A(\tau_z - i\tau_x s_x) + i\lambda_R s_y, \quad (146)$$

$$t_y = A(\tau_z - i\tau_x s_y) - i\lambda_R s_x. \quad (147)$$

The first terms in Eqs. (146) and (147) are the original nearest-neighbour hopping, whereas the terms proportional to λ_R describe genuine Rashba spin mixing.

We use the Fourier convention

$$c_{\mathbf{r}} = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}} c_{\mathbf{k}}. \quad (148)$$

A hopping matrix t_{δ} from $\mathbf{r}$ to $\mathbf{r} + \delta$ then contributes

$$t_{\delta} e^{i\mathbf{k} \cdot \delta} + t_{\delta}^{\dagger} e^{-i\mathbf{k} \cdot \delta} \quad (149)$$

to the Bloch Hamiltonian.

For the x bond,

$$\begin{aligned} t_x e^{ik_x} + t_x^{\dagger} e^{-ik_x} \\ = 2A \cos k_x \tau_z + 2A \sin k_x \tau_x s_x - 2\lambda_R \sin k_x s_y, \end{aligned} \quad (150)$$

while the y bond gives

$$\begin{aligned} t_y e^{ik_y} + t_y^{\dagger} e^{-ik_y} \\ = 2A \cos k_y \tau_z + 2A \sin k_y \tau_x s_y + 2\lambda_R \sin k_y s_x. \end{aligned} \quad (151)$$

Hence

$$\begin{aligned} H(\mathbf{k}) = d(\mathbf{k})\tau_z + 2A(\sin k_x \tau_x s_x + \sin k_y \tau_x s_y) \\ + 2\lambda_R(\sin k_y s_x - \sin k_x s_y), \end{aligned} \quad (152)$$

with

$$d(\mathbf{k}) = M + 2A(\cos k_x + \cos k_y). \quad (153)$$

Thus Eqs. (145) and (152) describe exactly the same Hamiltonian. Momentum space is being used here only as a convenient representation of the clean real-space model.

The Rashba term

$$H_R(\mathbf{k}) = 2\lambda_R(\sin k_y s_x - \sin k_x s_y) \quad (154)$$

preserves time reversal,

$$TH_R(\mathbf{k})T^{-1} = H_R(-\mathbf{k}), \quad (155)$$

but breaks the exact S_0 symmetry,

$$[H_R, S_0] \neq 0, \quad S_0 = \tau_z s_z. \quad (156)$$

This is therefore a genuine physical perturbation rather than a local change of basis.

XIX. CHOOSING S IN THE PRESENCE OF RASHBA SPIN MIXING

The unperturbed two-dimensional model has the exact local time-reversal-odd grading

$$S_0 = \tau_z s_z, \quad (157)$$

with

$$[S_0, H_0] = 0. \quad (158)$$

The aim is to understand how this operator should be modified after adding a genuine spin-mixing perturbation.

For the clean square lattice, consider

$$H(\mathbf{k}) = d(\mathbf{k})\tau_z + 2A(\sin k_x \tau_x s_x + \sin k_y \tau_x s_y) + 2\lambda_R(\sin k_y s_x - \sin k_x s_y), \quad (159)$$

where

$$d(\mathbf{k}) = M + 2A(\cos k_x + \cos k_y). \quad (160)$$

The last term is the Rashba perturbation,

$$H_R = 2\lambda_R(\sin k_y s_x - \sin k_x s_y). \quad (161)$$

It preserves time reversal but breaks the exact S_0 symmetry.

The form of the appropriate deformation of S_0 can be read directly from the commutators. Using

$$[s_z, s_x] = 2is_y, \quad [s_z, s_y] = -2is_x, \quad (162)$$

gives

$$[S_0, H_R] = 4i\lambda_R(\sin k_x \tau_z s_x + \sin k_y \tau_z s_y). \quad (163)$$

Now consider τ_y . Its commutator with the original spin-orbit hopping is

$$\begin{aligned} & [\tau_y, 2A(\sin k_x \tau_x s_x + \sin k_y \tau_x s_y)] \\ &= -4iA(\sin k_x \tau_z s_x + \sin k_y \tau_z s_y). \end{aligned} \quad (164)$$

The two commutators therefore have exactly the same matrix structure and opposite signs:

$$\boxed{[S_0, H_R] = -\frac{\lambda_R}{A}[\tau_y, H_{SO}].} \quad (165)$$

This singles out τ_y as the natural direction in which to deform S_0 .

This can also be formulated as a first-order deformation problem. Suppose that for weak Rashba coupling

$$S = S_0 + \lambda_R \delta S + O(\lambda_R^2). \quad (166)$$

To first order in λ_R ,

$$[S, H_0 + H_R] = [S_0, H_R] + \lambda_R[\delta S, H_0] + O(\lambda_R^2), \quad (167)$$

where $[S_0, H_0] = 0$ has been used. Restricting momentarily to the spin-orbit hopping part of H_0 , Eq. (??) shows that the choice

$$\delta S \propto \tau_y \quad (168)$$

is precisely the local direction required to cancel the Rashba-induced commutator. Thus τ_y is not an arbitrary variational generator: it is the leading deformation direction selected by the perturbation.

The deformation must, however, remain within the space of admissible operators entering the marker. This requires in particular

$$S^\dagger = S, \quad S^2 = 1, \quad \text{Tr } S = 0, \quad TST^{-1} = -S. \quad (169)$$

The two generators

$$S_0 = \tau_z s_z, \quad S_1 = \tau_y \quad (170)$$

have exactly the required algebraic properties:

$$S_0^\dagger = S_0, \quad S_1^\dagger = S_1, \quad (171)$$

$$S_0^2 = S_1^2 = 1, \quad (172)$$

and

$$\{S_0, S_1\} = \{\tau_z s_z, \tau_y\} = 0. \quad (173)$$

Therefore any normalized linear combination

$$S = aS_0 + bS_1 \quad (174)$$

satisfies

$$\begin{aligned} S^2 &= a^2 S_0^2 + b^2 S_1^2 + ab\{S_0, S_1\} \\ &= (a^2 + b^2)1. \end{aligned} \quad (175)$$

The involution condition $S^2 = 1$ hence reduces simply to

$$a^2 + b^2 = 1. \quad (176)$$

It is therefore natural to write

$$a = \cos \theta, \quad b = \sin \theta, \quad (177)$$

which gives

$$\boxed{S(\theta) = \cos \theta \tau_z s_z + \sin \theta \tau_y.} \quad (178)$$

Thus the Rashba perturbation does not motivate a completely arbitrary search for S . It naturally reduces the problem to finding the best angle θ in Eq. (??).

A. Why the hopping-preferred angle is not the optimum

For the hopping part of the Hamiltonian,

$$H_{\text{hop}} = H_{\text{SO}} + H_R, \quad (179)$$

Eqs. (163) and (164) give

$$[S(\theta), H_{\text{hop}}] = 4i (\lambda_R \cos \theta - A \sin \theta) (\sin k_x \tau_z s_x + \sin k_y \tau_z s_y). \quad (180)$$

The hopping therefore commutes with $S(\theta)$ at

$$\boxed{\theta_{\text{hop}} = \arctan \frac{\lambda_R}{A}.} \quad (181)$$

However, this is not a symmetry of the full Hamiltonian. The mass term satisfies

$$[S_0, d(\mathbf{k})\tau_z] = 0, \quad (182)$$

whereas

$$[\tau_y, d(\mathbf{k})\tau_z] = 2id(\mathbf{k})\tau_x. \quad (183)$$

Hence the mass term favours $\theta = 0$, while the hopping favours $\theta = \theta_{\text{hop}}$.

The appropriate angle should therefore not be obtained by demanding $[H, S] = 0$. The marker only requires

$$\tilde{S} = \{\rho, S\} - S \quad (184)$$

to remain gapped. The relevant problem is instead

$$\boxed{\theta_{\text{opt}} = \operatorname{argmax}_{\theta} \Delta_{\tilde{S}}(\theta),} \quad (185)$$

where

$$\Delta_{\tilde{S}} = \min |\operatorname{spec} \tilde{S}|. \quad (186)$$

B. Auxiliary gap in the clean system

The clean system is most easily analysed in momentum space. This is only a way of solving the S -selection problem analytically: $S(\theta)$ itself is an on-site operator and can subsequently be used unchanged in real space.

Define

$$r = \sqrt{\sin^2 k_x + \sin^2 k_y}, \quad (187)$$

and, for $r \neq 0$,

$$s_{\parallel} = \frac{\sin k_x s_x + \sin k_y s_y}{r}, \quad (188)$$

$$s_{\perp} = \frac{\sin k_y s_x - \sin k_x s_y}{r}. \quad (189)$$

These are orthogonal spin directions,

$$s_{\parallel}^2 = s_{\perp}^2 = 1, \quad \{s_{\parallel}, s_{\perp}\} = 0. \quad (190)$$

The Hamiltonian becomes

$$\boxed{H = d\tau_z + a\tau_x s_{\parallel} + b s_{\perp}}, \quad (191)$$

with

$$a = 2Ar, \quad b = 2\lambda_R r. \quad (192)$$

In particular,

$$\boxed{\frac{b}{a} = \frac{\lambda_R}{A}}, \quad (193)$$

which is independent of momentum.

The positive energies of Eq. (191) are

$$E_{\pm} = \sqrt{a^2 + (d \pm b)^2}. \quad (194)$$

At half filling,

$$Q = 1 - 2\rho = \text{sgn } H, \quad (195)$$

so that

$$\boxed{\tilde{S} = -\frac{1}{2}\{Q, S\}}. \quad (196)$$

To evaluate the spectrum of $\tilde{S}$, it is convenient to first determine the flattened Hamiltonian Q . At fixed momentum,

$$H = d\tau_z + a\tau_x s_{\parallel} + b s_{\perp}, \quad (197)$$

with

$$s_{\parallel}^2 = s_{\perp}^2 = 1, \quad \{s_{\parallel}, s_{\perp}\} = 0. \quad (198)$$

Squaring the Hamiltonian gives

$$H^2 = a^2 + d^2 + b^2 + 2db\tau_z s_{\perp}. \quad (199)$$

Introducing

$$K = \tau_z s_{\perp}, \quad K^2 = 1, \quad (200)$$

we note that K commutes with the Hamiltonian,

$$[K, H] = 0. \quad (201)$$

The eigenstates of H can therefore be chosen to have definite $K = \pm 1$. In a sector with $K = k$, $k = \pm 1$, the squared Hamiltonian reduces to

$$H^2 = a^2 + d^2 + b^2 + 2dbk. \quad (202)$$

Hence the two positive energy magnitudes are

$$E_{\pm} = \sqrt{a^2 + (d \pm b)^2}. \quad (203)$$

The corresponding Hamiltonian eigenvalues are $\pm E_+$ and $\pm E_-$. With the corresponding projectors

$$\Pi_{\pm} = \frac{1 \pm K}{2}, \quad (204)$$

we may write

$$|H|^{-1} = \frac{\Pi_+}{E_+} + \frac{\Pi_-}{E_-}, \quad (205)$$

and hence

$$Q = H \left(\frac{\Pi_+}{E_+} + \frac{\Pi_-}{E_-} \right). \quad (206)$$

We now evaluate the anticommutator with

$$S(\theta) = \cos \theta \tau_z s_z + \sin \theta \tau_y. \quad (207)$$

At each fixed momentum we can rotate the spin basis within the xy plane so that

$$s_{\parallel} = s_x, \quad s_{\perp} = -s_y. \quad (208)$$

This rotation leaves s_z , and therefore $S(\theta)$, unchanged.

Writing

$$c = \cos \theta, \quad s = \sin \theta, \quad (209)$$

a direct evaluation gives

$$\frac{1}{2}\{Q, S\} = \begin{pmatrix} u & 0 & 0 & v \\ 0 & -u & -v & 0 \\ 0 & -v & u & 0 \\ v & 0 & 0 & -u \end{pmatrix}, \quad (210)$$

where

$$u = \frac{1}{2} \left[\frac{c(d+b) - as}{E_+} + \frac{c(d-b) + as}{E_-} \right], \quad (211)$$

$$v = \frac{1}{2} \left[\frac{ac + s(d+b)}{E_+} + \frac{ac - s(d-b)}{E_-} \right]. \quad (212)$$

The important point is that this matrix squares to a multiple of the identity,

$$\left(\frac{1}{2}\{Q, S\} \right)^2 = (u^2 + v^2) \mathbb{K}_4. \quad (213)$$

Since

$$\tilde{S} = -\frac{1}{2}\{Q, S\}, \quad (214)$$

we therefore have

$$\boxed{\tilde{S}^2 = \delta_S^2 \mathbb{I}_4}, \quad (215)$$

with

$$\delta_S = \sqrt{u^2 + v^2}. \quad (216)$$

Thus all four eigenvalues have the same magnitude. Since $\tilde{S}$ is time-reversal odd, they occur in opposite-sign pairs, so that

$$\boxed{\text{spec } \tilde{S}(\mathbf{k}, \theta) = \{-\delta_S, -\delta_S, +\delta_S, +\delta_S\}}. \quad (217)$$

The expression for δ_S simplifies to

$$\delta_S^2 = \frac{1}{2} \left[1 + \frac{(a^2 + d^2 - b^2) \cos 2\theta + 2ab \sin 2\theta}{E_+ E_-} \right]. \quad (218)$$

Using

$$(E_+ E_-)^2 = (a^2 + d^2 - b^2)^2 + 4a^2 b^2, \quad (219)$$

we define a momentum-dependent angle $\vartheta(\mathbf{k})$ by

$$\boxed{\cos 2\vartheta(\mathbf{k}) = \frac{a^2 + d^2 - b^2}{E_+ E_-}}, \quad (220)$$

$$\boxed{\sin 2\vartheta(\mathbf{k}) = \frac{2ab}{E_+ E_-}}. \quad (221)$$

Then

$$\delta_S^2 = \frac{1}{2} [1 + \cos 2\theta \cos 2\vartheta + \sin 2\theta \sin 2\vartheta] \quad (222)$$

$$= \frac{1}{2} [1 + \cos 2(\theta - \vartheta)] \quad (223)$$

$$= \cos^2(\theta - \vartheta). \quad (224)$$

Therefore

$$\boxed{\delta_S = |\cos[\theta - \vartheta(\mathbf{k})]|}. \quad (225)$$

Finally, using

$$\tan \vartheta = \frac{\sin 2\vartheta}{1 + \cos 2\vartheta}, \quad (226)$$

we obtain

$$\boxed{\tan \vartheta(\mathbf{k}) = \frac{2ab}{E_+ E_- + a^2 + d^2 - b^2}}. \quad (227)$$

Thus the problem has a simple interpretation: different momenta favour slightly different orientations of S , parametrized by $\vartheta(\mathbf{k})$, and θ must be chosen to remain as far as possible from closing the $\tilde{S}$ gap for any momentum.

C. Range of the preferred angle

For the present model,

$$0 \leq \vartheta(\mathbf{k}) \leq \arctan \frac{\lambda_R}{A}. \quad (228)$$

This follows directly from Eq. (??). Indeed,

$$(E_+ E_-)^2 - (a^2 + b^2 - d^2)^2 = 4a^2 d^2 \geq 0, \quad (229)$$

and therefore

$$E_+ E_- \geq a^2 + b^2 - d^2. \quad (230)$$

The denominator of Eq. (??) consequently satisfies

$$E_+ E_- + a^2 + d^2 - b^2 \geq 2a^2. \quad (231)$$

Hence

$$\tan \vartheta \leq \frac{b}{a} = \frac{\lambda_R}{A}. \quad (232)$$

Defining

$$\theta_{\text{hop}} = \arctan \frac{\lambda_R}{A}, \quad (233)$$

gives

$$\boxed{0 \leq \vartheta(\mathbf{k}) \leq \theta_{\text{hop}}.} \quad (234)$$

For the parameters considered here, both limiting values are realised. At a time-reversal-invariant momentum,

$$\sin k_x = \sin k_y = 0, \quad (235)$$

so $a = b = 0$ and

$$\vartheta = 0. \quad (236)$$

At a momentum where

$$d(\mathbf{k}) = 0, \quad (237)$$

Eq. (??) gives

$$\tan \vartheta = \frac{b}{a} = \frac{\lambda_R}{A}, \quad (238)$$

and therefore

$$\vartheta = \theta_{\text{hop}}. \quad (239)$$

For $M = -2$ and $A = 1$, for example,

$$(k_x, k_y) = \left(\frac{\pi}{2}, 0\right) \quad (240)$$

has $d(\mathbf{k}) = 0$.

D. Optimal angle

For a fixed choice of θ , the auxiliary gap depends on momentum as

$$\Delta_{\tilde{S}}(\mathbf{k}, \theta) = |\cos[\theta - \vartheta(\mathbf{k})]|. \quad (241)$$

In the clean system the full operator $\tilde{S}$ is block diagonal in momentum, so its spectral gap is set by the momentum at which this quantity is smallest:

$$\Delta_{\tilde{S}}(\theta) = \min_{\mathbf{k}} \Delta_{\tilde{S}}(\mathbf{k}, \theta). \quad (242)$$

This is analogous to an ordinary band gap: if the gap closes at even one momentum, then the full operator $\tilde{S}$ is gapless. We found that

$$0 \leq \vartheta(\mathbf{k}) \leq \theta_{\text{hop}}, \quad (243)$$

and that both limiting values are realised somewhere in the Brillouin zone. Thus, for a given θ with $0 \leq \theta \leq \theta_{\text{hop}}$, the momenta with $\vartheta = 0$ and $\vartheta = \theta_{\text{hop}}$ give

$$\Delta_{\tilde{S}} = \cos \theta \quad (244)$$

and

$$\Delta_{\tilde{S}} = \cos(\theta_{\text{hop}} - \theta), \quad (245)$$

respectively.

All other momenta have

$$0 < \vartheta(\mathbf{k}) < \theta_{\text{hop}}, \quad (246)$$

and therefore lie between these two limiting orientations. Since $\cos x$ decreases with x for $0 \leq x \leq \pi/2$, the smallest gap is produced by whichever endpoint is farther from the chosen angle θ . Hence

$$\Delta_{\tilde{S}}(\theta) = \min \{ \cos \theta, \cos(\theta_{\text{hop}} - \theta) \}. \quad (247)$$

The problem is therefore to choose θ so that this worst-case gap is as large as possible:

$$\theta_{\text{opt}} = \underset{\theta}{\operatorname{argmax}} \Delta_{\tilde{S}}(\theta). \quad (248)$$

The first term,

$$\cos \theta, \quad (249)$$

decreases with θ , while the second,

$$\cos(\theta_{\text{hop}} - \theta), \quad (250)$$

increases. The maximum of their minimum is therefore reached when the two are equal,

$$\cos \theta = \cos(\theta_{\text{hop}} - \theta). \quad (251)$$

Within the interval $0 \leq \theta \leq \theta_{\text{hop}}$ this gives

$$\theta = \theta_{\text{hop}} - \theta, \quad (252)$$

and therefore

$$\theta_{\text{opt}} = \frac{\theta_{\text{hop}}}{2} = \frac{1}{2} \arctan \frac{\lambda_R}{A}. \quad (253)$$

Substituting the optimal angle back into $S(\theta)$ gives the optimized local grading

$$S_{\text{opt}} = \cos \left[\frac{1}{2} \arctan \frac{\lambda_R}{A} \right] \tau_z s_z + \sin \left[\frac{1}{2} \arctan \frac{\lambda_R}{A} \right] \tau_y. \quad (254)$$

At the optimal angle the two limiting values of the auxiliary gap are equal. Therefore

$$\Delta_{\tilde{S}}[S_{\text{opt}}] = \cos \theta_{\text{opt}} = \cos \left[\frac{1}{2} \arctan \frac{\lambda_R}{A} \right]. \quad (255)$$

Using the half-angle identity,

$$\cos \frac{\phi}{2} = \sqrt{\frac{1 + \cos \phi}{2}}, \quad (256)$$

together with

$$\cos \theta_{\text{hop}} = \frac{A}{\sqrt{A^2 + \lambda_R^2}}, \quad (257)$$

we obtain

$$\Delta_{\tilde{S}}[S_{\text{opt}}] = \sqrt{\frac{1}{2} \left(1 + \frac{A}{\sqrt{A^2 + \lambda_R^2}} \right)}. \quad (258)$$

E. Auxiliary gap for the original grading S_0

Before optimizing the grading, it is useful to determine how the original choice

$$S_0 = \tau_z s_z \quad (259)$$

behaves in the presence of Rashba spin mixing.

The general result obtained above is

$$\Delta_{\tilde{S}}(\mathbf{k}, \theta) = |\cos[\theta - \vartheta(\mathbf{k})]|, \quad (260)$$

where

$$0 \leq \vartheta(\mathbf{k}) \leq \theta_{\text{hop}}, \quad (261)$$

with

$$\theta_{\text{hop}} = \arctan \frac{\lambda_R}{A}. \quad (262)$$

The original grading S_0 corresponds to $\theta = 0$. Therefore

$$\Delta_{\tilde{S}}(\mathbf{k}, S_0) = \cos \vartheta(\mathbf{k}). \quad (263)$$

The gap of the full clean system is the minimum over momentum,

$$\Delta_{\tilde{S}}[S_0] = \min_{\mathbf{k}} \cos \vartheta(\mathbf{k}). \quad (264)$$

Since $\cos \vartheta$ decreases on the interval considered here, the smallest value occurs at the largest value of ϑ ,

$$\vartheta = \theta_{\text{hop}}. \quad (265)$$

This endpoint is attained at momenta satisfying $d(\mathbf{k}) = 0$. Hence

$$\Delta_{\tilde{S}}[S_0] = \cos \theta_{\text{hop}}. \quad (266)$$

Using

$$\tan \theta_{\text{hop}} = \frac{\lambda_R}{A}, \quad (267)$$

we have

$$\cos \theta_{\text{hop}} = \frac{1}{\sqrt{1 + \tan^2 \theta_{\text{hop}}}} = \frac{A}{\sqrt{A^2 + \lambda_R^2}}. \quad (268)$$

Thus

$$\boxed{\Delta_{\tilde{S}}[S_0] = \frac{A}{\sqrt{A^2 + \lambda_R^2}}.} \quad (269)$$

The original grading therefore remains admissible for every finite λ_R , but its auxiliary gap decreases continuously as the Rashba coupling is increased. Thus both choices remain gapped for finite λ_R , but the optimized grading has the larger auxiliary gap,

$$\Delta_{\tilde{S}}[S_{\text{opt}}] > \Delta_{\tilde{S}}[S_0] \quad (\lambda_R > 0). \quad (270)$$

The optimization therefore does not define a different topological sector; it gives a more robust separation of the positive and negative $\tilde{S}$ sectors.

XX. GENERALIZATION TO ORTHOGONAL SPIN TEXTURES

The previous calculation does not rely on the particular momentum dependence $\sin k_x$ and $\sin k_y$ of the Rashba model. Consider the more general class of Hamiltonians

$$\boxed{H(\mathbf{k}) = d(\mathbf{k})\tau_z + \tau_x [g_x(\mathbf{k})s_x + g_y(\mathbf{k})s_y] + \gamma [g_y(\mathbf{k})s_x - g_x(\mathbf{k})s_y],} \quad (271)$$

where $d(\mathbf{k})$, $g_x(\mathbf{k})$, and $g_y(\mathbf{k})$ are real functions and γ is a momentum-independent real parameter. For a time-reversal-invariant model one may take

$$d(-\mathbf{k}) = d(\mathbf{k}), \quad g_i(-\mathbf{k}) = -g_i(\mathbf{k}). \quad (272)$$

Define

$$g(\mathbf{k}) = \sqrt{g_x(\mathbf{k})^2 + g_y(\mathbf{k})^2}, \quad (273)$$

and, whenever $g(\mathbf{k}) \neq 0$, the normalized spin operators

$$s_{\parallel}(\mathbf{k}) = \frac{g_x(\mathbf{k})s_x + g_y(\mathbf{k})s_y}{g(\mathbf{k})}, \quad (274)$$

$$s_{\perp}(\mathbf{k}) = \frac{g_y(\mathbf{k})s_x - g_x(\mathbf{k})s_y}{g(\mathbf{k})}. \quad (275)$$

They satisfy

$$s_{\parallel}^2 = s_{\perp}^2 = 1, \quad \{s_{\parallel}, s_{\perp}\} = 0. \quad (276)$$

The Hamiltonian can therefore be written as

$$\boxed{H(\mathbf{k}) = d(\mathbf{k})\tau_z + g(\mathbf{k})\tau_x s_{\parallel}(\mathbf{k}) + \gamma g(\mathbf{k})s_{\perp}(\mathbf{k}).} \quad (277)$$

This has precisely the form considered in the previous subsection,

$$H = d\tau_z + a\tau_x s_{\parallel} + b s_{\perp}, \quad (278)$$

with the identification

$$\boxed{a(\mathbf{k}) = g(\mathbf{k}), \quad b(\mathbf{k}) = \gamma g(\mathbf{k}).} \quad (279)$$

In particular,

$$\boxed{\frac{b(\mathbf{k})}{a(\mathbf{k})} = \gamma} \quad (280)$$

is independent of momentum.

The result derived above for the auxiliary spectrum therefore applies without modification. For

$$S(\theta) = \cos \theta \tau_z s_z + \sin \theta \tau_y, \quad (281)$$

the eigenvalues of $\tilde{S}$ have magnitude

$$\boxed{\delta_S(\mathbf{k}, \theta) = |\cos[\theta - \vartheta(\mathbf{k})]|}, \quad (282)$$

where

$$\tan \vartheta(\mathbf{k}) = \frac{2\gamma g(\mathbf{k})^2}{E_+ E_- + [1 - \gamma^2]g(\mathbf{k})^2 + d(\mathbf{k})^2}, \quad (283)$$

and

$$E_{\pm} = \sqrt{g(\mathbf{k})^2 + [d(\mathbf{k}) \pm \gamma g(\mathbf{k})]^2}. \quad (284)$$

We next determine the range of the momentum-dependent angle $\vartheta(\mathbf{k})$. The identity

$$(E_+ E_-)^2 - [(1 + \gamma^2)g^2 - d^2]^2 = 4g^2 d^2 \geq 0 \quad (285)$$

implies

$$E_+ E_- \geq (1 + \gamma^2)g^2 - d^2. \quad (286)$$

The denominator of Eq. (283) therefore satisfies

$$\begin{aligned} E_+ E_- + (1 - \gamma^2)g^2 + d^2 \\ \geq (1 + \gamma^2)g^2 - d^2 + (1 - \gamma^2)g^2 + d^2 \\ = 2g^2. \end{aligned} \quad (287)$$

For $\gamma \geq 0$ this gives

$$0 \leq \tan \vartheta(\mathbf{k}) \leq \gamma, \quad (288)$$

and hence

$$\boxed{0 \leq \vartheta(\mathbf{k}) \leq \arctan \gamma.} \quad (289)$$

The lower endpoint is realized at momenta where $g(\mathbf{k}) = 0$, provided the physical gap remains open there. At such a point,

$$\vartheta(\mathbf{k}) = 0. \quad (290)$$

The upper endpoint is realized at momenta where $d(\mathbf{k}) = 0$ and $g(\mathbf{k}) \neq 0$. In that case

$$E_+ = E_- = g\sqrt{1 + \gamma^2}, \quad (291)$$

and Eq. (283) reduces to

$$\tan \vartheta = \gamma. \quad (292)$$

Thus

$$\vartheta = \arctan \gamma. \quad (293)$$

If the Brillouin zone contains momenta realizing both endpoints, the global auxiliary gap is therefore

$$\Delta_{\tilde{S}}(\theta) = \min \{ \cos \theta, \cos[\arctan \gamma - \theta] \}. \quad (294)$$

The first term decreases with θ , while the second increases, so the maximum of the minimum is obtained when they are equal. This gives

$$\theta_{\text{opt}} = \frac{1}{2} \arctan \gamma. \quad (295)$$

The corresponding optimized local grading is

$$S_{\text{opt}} = \cos \left(\frac{1}{2} \arctan \gamma \right) \tau_z s_z + \sin \left(\frac{1}{2} \arctan \gamma \right) \tau_y. \quad (296)$$

The Rashba model considered above is recovered by choosing

$$g_x(\mathbf{k}) = 2A \sin k_x, \quad g_y(\mathbf{k}) = 2A \sin k_y, \quad \gamma = \frac{\lambda_R}{A}. \quad (297)$$

Equation (295) then reduces to

$$\theta_{\text{opt}} = \frac{1}{2} \arctan \frac{\lambda_R}{A}. \quad (298)$$

Thus the half-angle result does not depend on the specific square-lattice momentum dependence. It follows more generally when the spin-mixing texture is pointwise orthogonal to the orbital-spin texture and has a momentum-independent relative magnitude.

XXI. EXTENSION TO AN AMORPHOUS STRUCTURE

The clean calculation above motivates a local grading for a translation-invariant BHZ model with spin mixing. We now consider whether the same construction remains meaningful when translation symmetry is removed by structural disorder.

We follow the random-lattice construction of Agarwala and Shenoy [2]. In this construction, the sites are placed at random positions $\mathbf{r}_i$ in a finite region of space, and pairs of sites within a finite hopping range are connected. A bond between two sites i and j is characterized by the displacement

$$\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j = r_{ij} \begin{pmatrix} \cos \phi_{ij} \\ \sin \phi_{ij} \end{pmatrix}. \quad (299)$$

Thus, in contrast to the square lattice, the bond angle ϕ_{ij} is not restricted to the coordinate directions.

Agarwala and Shenoy write a generic random-lattice hopping in the factorized form

$$t_{\alpha\beta}(\mathbf{r}) = t(r) T_{\alpha\beta}(\hat{\mathbf{r}}), \quad (300)$$

where, in their words, “the distance dependence of the hopping is captured by $t(r)$ ”, while the matrix $T(\hat{\mathbf{r}})$ contains the orbital and angular dependence [2]. The radial hopping in their construction is chosen to decay exponentially and to vanish beyond a finite cutoff,

$$t(r) = C \Theta(R - r) e^{-r/a}. \quad (301)$$

The precise radial profile is not important for the discussion below; we denote it more generally by $f(r)$.

We construct an amorphous continuation of the BHZ–Rashba model by retaining the same local orbital and spin degrees of freedom at every site, while replacing the square-lattice bond directions by the actual directions of the random bonds. For each bond, define the two in-plane spin operators

$$s_{\parallel}(\phi_{ij}) = \cos \phi_{ij} s_x + \sin \phi_{ij} s_y, \quad (302)$$

$$s_{\perp}(\phi_{ij}) = \sin \phi_{ij} s_x - \cos \phi_{ij} s_y. \quad (303)$$

These operators satisfy

$$s_{\parallel}^2 = s_{\perp}^2 = 1, \quad \{s_{\parallel}, s_{\perp}\} = 0. \quad (304)$$

Thus $s_{\parallel}$ points along the bond direction in spin space, while $s_{\perp}$ is the corresponding in-plane perpendicular direction.

A natural hopping matrix on the bond ij is then

$$\boxed{T_{ij} = f(r_{ij}) [A\tau_z - iA\tau_x s_{\parallel}(\phi_{ij}) - i\lambda_R s_{\perp}(\phi_{ij})]}. \quad (305)$$

The full Hamiltonian is

$$H_{\text{amor}} = M \sum_i c_i^{\dagger} \tau_z c_i + \sum_{i < j} \left[c_i^{\dagger} T_{ij} c_j + \text{h.c.} \right], \quad (306)$$

where the second sum runs over the bonds included in the random lattice.

This construction reduces to the square-lattice model for bonds along the coordinate axes. For a bond along the positive x direction, $\phi = 0$, so that

$$s_{\parallel} = s_x, \quad s_{\perp} = -s_y, \quad (307)$$

and Eq. (305) becomes

$$T_x = f(r) [A\tau_z - iA\tau_x s_x + i\lambda_R s_y]. \quad (308)$$

For a bond along the positive y direction, $\phi = \pi/2$, giving

$$s_{\parallel} = s_y, \quad s_{\perp} = s_x, \quad (309)$$

and therefore

$$T_y = f(r) [A\tau_z - iA\tau_x s_y - i\lambda_R s_x]. \quad (310)$$

Up to the common radial factor, these are precisely the square-lattice hopping matrices used above.

The directional dependence also guarantees

$$T(-\mathbf{r}) = T(\mathbf{r})^{\dagger}, \quad (311)$$

since both $s_{\parallel}$ and $s_{\perp}$ change sign under $\phi \rightarrow \phi + \pi$. The bond Hamiltonian is therefore Hermitian. Moreover, the construction preserves time-reversal symmetry: both in-plane spin operators are odd under time reversal, while i changes sign, leaving the spin-dependent hopping terms invariant.

We next ask whether the local family of grading operators introduced for the crystalline model remains natural in the amorphous system. Consider again

$$\boxed{S(\theta) = \cos \theta S_0 + \sin \theta \tau_y, \quad S_0 = \tau_z s_z}. \quad (312)$$

In real space this is an onsite operator,

$$S(\theta) = \sum_i |i\rangle \langle i| \otimes [\cos \theta \tau_z s_z + \sin \theta \tau_y]. \quad (313)$$

It therefore remains well defined without translation symmetry.

Importantly, the algebra that motivated the two components of $S(\theta)$ survives for an arbitrary bond direction. From the definitions above,

$$[s_z, s_{\perp}] = 2is_{\parallel}. \quad (314)$$

It follows that

$$[S_0, s_\perp] = 2i\tau_z s_\parallel. \quad (315)$$

On the other hand,

$$[\tau_y, \tau_x s_\parallel] = -2i\tau_z s_\parallel. \quad (316)$$

The two commutators therefore generate exactly the same local matrix structure, with opposite signs.

To make this explicit, separate the spin-dependent part of a bond hopping into

$$T_{ij}^{\text{SO}} = -iAf(r_{ij})\tau_x s_\parallel(\phi_{ij}), \quad (317)$$

$$T_{ij}^R = -i\lambda_R f(r_{ij})s_\perp(\phi_{ij}). \quad (318)$$

Then

$$[S_0, T_{ij}^R] = 2\lambda_R f(r_{ij})\tau_z s_\parallel(\phi_{ij}), \quad (319)$$

$$[\tau_y, T_{ij}^{\text{SO}}] = -2Af(r_{ij})\tau_z s_\parallel(\phi_{ij}), \quad (320)$$

and hence, bond by bond,

$$\boxed{[S_0, T_{ij}^R] = -\frac{\lambda_R}{A}[\tau_y, T_{ij}^{\text{SO}}].} \quad (321)$$

This relation does not depend on the bond angle ϕ_{ij} or the bond length r_{ij} . Consequently it survives the sum over an arbitrary realization of the amorphous lattice,

$$\boxed{[S_0, H_R] = -\frac{\lambda_R}{A}[\tau_y, H_{\text{SO}}],} \quad (322)$$

provided that the BHZ spin-orbit and Rashba hopping channels have the same radial envelope $f(r)$ and a bond-independent ratio λ_R/A .

The spin-dependent hopping part therefore gives

$$\begin{aligned} [S(\theta), H_{\text{SO}} + H_R] \\ = \cos\theta [S_0, H_R] + \sin\theta [\tau_y, H_{\text{SO}}], \end{aligned} \quad (323)$$

where we used

$$[S_0, H_{\text{SO}}] = 0, \quad [\tau_y, H_R] = 0. \quad (324)$$

Using Eq. (322), this becomes proportional to

$$\lambda_R \cos\theta - A \sin\theta. \quad (325)$$

Thus the spin-dependent hopping part is exactly conserved by the grading angle

$$\boxed{\theta_{\text{hop}} = \arctan \frac{\lambda_R}{A},} \quad (326)$$

independently of the random positions and bond directions.

The remaining terms proportional to τ_z behave differently. This includes both the onsite mass term and the τ_z component of the hopping in Eq. (305). Since

$$[S_0, \tau_z] = 0, \quad [\tau_y, \tau_z] = 2i\tau_x, \quad (327)$$

these terms favour $\theta = 0$. The amorphous model therefore retains the same competition between two local matrix structures as the crystalline model:

$$\begin{aligned} \tau_z\text{-type terms} &\longrightarrow \theta = 0, \\ \text{spin-dependent hopping} &\longrightarrow \theta = \arctan \frac{\lambda_R}{A}. \end{aligned} \quad (328)$$

What does not carry over directly is the crystalline derivation of the midpoint angle. In the translation-invariant model, the Hamiltonian could be decomposed into independent momenta, and the auxiliary gap took the form

$$\Delta_{\tilde{S}}(\mathbf{k}, \theta) = |\cos[\theta - \vartheta(\mathbf{k})]|. \quad (329)$$

The range of $\vartheta(\mathbf{k})$ over the Brillouin zone then fixed the global maximin problem and gave

$$\theta_{\text{opt}}^{\text{clean}} = \frac{1}{2} \arctan \frac{\lambda_R}{A}. \quad (330)$$

There is no Brillouin zone, and therefore no corresponding momentum-resolved argument, on a generic random lattice.

The appropriate optimization in the amorphous system can instead be defined directly in real space. Let ρ_{amor} be the projector onto the occupied states of Eq. (306). For every local grading $S(\theta)$, define

$$\tilde{S}_{\text{amor}}(\theta) = \{\rho_{\text{amor}}, S(\theta)\} - S(\theta). \quad (331)$$

The corresponding auxiliary gap is

$$\Delta_{\tilde{S}}^{\text{amor}}(\theta) = \min_a \left| \lambda_a[\tilde{S}_{\text{amor}}(\theta)] \right|, \quad (332)$$

where λ_a denotes the eigenvalues of the full real-space operator. An optimal grading within the family Eq. (312) can then be defined by

$$\theta_{\text{opt}}^{\text{amor}} = \arg \max_{\theta} \Delta_{\tilde{S}}^{\text{amor}}(\theta). \quad (333)$$

Equations (321) and (322) show that structural disorder of the Agarwala type does not by itself introduce a new internal matrix direction into the spin-dependent hopping. Arbitrarily oriented bonds retain the same orthogonal pair $s_{\parallel}$ and $s_{\perp}$ and hence motivate the same local family $S(\theta)$ as in the crystalline problem. Structural disorder does, however, remove the momentum-space argument that fixed the exact optimal angle. The auxiliary gap Eq. (332) therefore provides the natural real-space criterion for determining whether the clean optimized grading remains appropriate in an amorphous realization.

XXII. RASHBA–DRESSELHAUS SPIN MIXING

The pure Rashba model considered above belongs to a particularly simple class of spin-mixing Hamiltonians. It is useful to understand which parts of the construction survive once this special structure is broken. To this end, consider first

$$H_{\text{BHZ+R}}(\mathbf{k}) = d(\mathbf{k})\tau_z + 2A \sin k_x \tau_x s_x + 2A \sin k_y \tau_x s_y + 2\lambda_R (\sin k_y s_x - \sin k_x s_y), \quad (334)$$

where

$$d(\mathbf{k}) = M + 2A(\cos k_x + \cos k_y). \quad (335)$$

Introducing

$$\mathbf{g}(\mathbf{k}) = 2A \begin{pmatrix} \sin k_x \\ \sin k_y \end{pmatrix}, \quad \mathbf{h}_R(\mathbf{k}) = 2\lambda_R \begin{pmatrix} \sin k_y \\ -\sin k_x \end{pmatrix}, \quad (336)$$

the spin-dependent part can be written as

$$H_{\text{spin}} = \tau_x \mathbf{g} \cdot \mathbf{s} + \mathbf{h}_R \cdot \mathbf{s}. \quad (337)$$

The two spin textures satisfy

$$\mathbf{g} \cdot \mathbf{h}_R = 0, \quad \frac{|\mathbf{h}_R|}{|\mathbf{g}|} = \frac{\lambda_R}{A}. \quad (338)$$

This pointwise orthogonality is the reason that the Rashba problem admits the simple conserved operator

$$K_R = \tau_z \hat{\mathbf{h}}_R \cdot \mathbf{s}, \quad [K_R, H_{\text{BHZ}+\text{R}}] = 0, \quad (339)$$

and ultimately allows the auxiliary spectrum associated with

$$S(\theta) = \cos \theta \tau_z s_z + \sin \theta \tau_y \quad (340)$$

to be obtained analytically.

The same Hamiltonian has the real-space form

$$H_{\text{BHZ}+\text{R}} = M \sum_{\mathbf{r}} c_{\mathbf{r}}^\dagger \tau_z c_{\mathbf{r}} + \sum_{\mathbf{r}} \left[c_{\mathbf{r}+\hat{x}}^\dagger t_x^{(R)} c_{\mathbf{r}} + c_{\mathbf{r}+\hat{y}}^\dagger t_y^{(R)} c_{\mathbf{r}} + \text{h.c.} \right], \quad (341)$$

with

$$t_x^{(R)} = A(\tau_z - \tau_x s_x) + \lambda_R s_y, \quad (342)$$

$$t_y^{(R)} = A(\tau_z - \tau_x s_y) - \lambda_R s_x. \quad (343)$$

Thus the Rashba spin direction is perpendicular to the BHZ spin-orbit direction on each nearest-neighbour bond.

A. Dresselhaus coupling.

A simple way to leave the orthogonal-texture class is to add the lattice form of the linear Dresselhaus coupling,

$$H_D(\mathbf{k}) = 2\lambda_D (\sin k_x s_x - \sin k_y s_y). \quad (344)$$

In the continuum, the Rashba and Dresselhaus interactions have the characteristic forms

$$H_R \sim k_y s_x - k_x s_y, \quad H_D \sim k_x s_x - k_y s_y, \quad (345)$$

respectively. Rashba coupling is conventionally associated with structural inversion asymmetry, whereas Dresselhaus coupling originates from bulk inversion asymmetry. Both preserve time reversal.

The full momentum-space Hamiltonian is

$$\begin{aligned} H_{RD}(\mathbf{k}) = & d(\mathbf{k})\tau_z + 2A \sin k_x \tau_x s_x + 2A \sin k_y \tau_x s_y \\ & + 2\lambda_R (\sin k_y s_x - \sin k_x s_y) \\ & + 2\lambda_D (\sin k_x s_x - \sin k_y s_y). \end{aligned} \quad (346)$$

In real space,

$$H_{RD} = M \sum_{\mathbf{r}} c_{\mathbf{r}}^\dagger \tau_z c_{\mathbf{r}} + \sum_{\mathbf{r}} \left[c_{\mathbf{r}+\hat{x}}^\dagger t_x^{(RD)} c_{\mathbf{r}} + c_{\mathbf{r}+\hat{y}}^\dagger t_y^{(RD)} c_{\mathbf{r}} + \text{h.c.} \right], \quad (347)$$

where

$$t_x^{(RD)} = A(\tau_z - \tau_x s_x) + \lambda_R s_y - \lambda_D s_x, \quad (348)$$

$$t_y^{(RD)} = A(\tau_z - \tau_x s_y) - \lambda_R s_x + \lambda_D s_y. \quad (349)$$

The distinction from pure Rashba coupling is already visible here. On an x bond, the BHZ and Dresselhaus terms both involve s_x , whereas the Rashba term involves s_y ; on a y bond, the BHZ and Dresselhaus terms both involve s_y , whereas the Rashba term involves s_x .

For later use, write Eq. (346) in the more general form

$$H = d\tau_z + \tau_x \mathbf{g} \cdot \mathbf{s} + \mathbf{h} \cdot \mathbf{s}, \quad (350)$$

with

$$\mathbf{g} = 2A \begin{pmatrix} p_x \\ p_y \end{pmatrix}, \quad \mathbf{h} = 2 \begin{pmatrix} \lambda_R p_y + \lambda_D p_x \\ -\lambda_R p_x - \lambda_D p_y \end{pmatrix}, \quad p_i = \sin k_i. \quad (351)$$

Unlike the pure Rashba case,

$$\boxed{\mathbf{g} \cdot \mathbf{h} = 4A\lambda_D(p_x^2 - p_y^2),} \quad (352)$$

which is nonzero at generic momentum. Consequently the simple operator $\tau_z \hat{\mathbf{h}} \cdot \mathbf{s}$ no longer commutes with the Hamiltonian. Indeed,

$$\left[\tau_z \hat{\mathbf{h}} \cdot \mathbf{s}, \tau_x \mathbf{g} \cdot \mathbf{s} \right] = 2i(\hat{\mathbf{h}} \cdot \mathbf{g})\tau_y. \quad (353)$$

Dresselhaus coupling therefore removes the simple two-sector decomposition used for the Rashba problem.

a. Generalized sector decomposition. Although the simple operator K_R is lost, the Hamiltonian retains enough algebraic structure to construct a more general conserved involution. Squaring Eq. (350) gives

$$H^2 = \alpha \mathcal{K} + 2B, \quad (354)$$

where

$$\alpha = d^2 + |\mathbf{g}|^2 + |\mathbf{h}|^2, \quad (355)$$

and

$$\boxed{B = d\tau_z \mathbf{h} \cdot \mathbf{s} + (\mathbf{g} \cdot \mathbf{h})\tau_x.} \quad (356)$$

The two terms in B anticommute, so that

$$B^2 = R^2 \mathcal{K}, \quad R^2 = d^2 |\mathbf{h}|^2 + (\mathbf{g} \cdot \mathbf{h})^2. \quad (357)$$

Moreover, Eq. (354) implies $[B, H] = 0$. Hence, whenever $R \neq 0$,

$$K_{\text{gen}} = \frac{B}{R} \quad (358)$$

satisfies

$$K_{\text{gen}}^2 = 1, \quad [K_{\text{gen}}, H] = 0. \quad (359)$$

The corresponding projectors are

$$\Pi_{\pm} = \frac{1 \pm K_{\text{gen}}}{2}, \quad (360)$$

and the two positive energy scales are

$$\boxed{E_{\pm} = \sqrt{\alpha \pm 2R}.} \quad (361)$$

The spectrally flattened Hamiltonian therefore takes the exact form

$$\boxed{Q = \text{sgn } H = H \left(\frac{\Pi_+}{E_+} + \frac{\Pi_-}{E_-} \right).} \quad (362)$$

The apparent singularity of $K_{\text{gen}} = B/R$ at $R = 0$ is not a singularity of Q : the two sectors become degenerate there and the expression above has a smooth limit as long as the physical energy gap remains open.

b. Auxiliary spectrum. The central question is whether the loss of the simple Rashba sector operator also destroys the one-angle grading Eq. (340). It does not.

To see this without diagonalizing a general 4×4 matrix, write

$$\frac{\Pi_+}{E_+} + \frac{\Pi_-}{E_-} = u \mathbb{1} + vB, \quad (363)$$

where

$$u = \frac{1}{2} \left(\frac{1}{E_+} + \frac{1}{E_-} \right), \quad v = \frac{1}{2R} \left(\frac{1}{E_+} - \frac{1}{E_-} \right). \quad (364)$$

Multiplication by H brings Q to the compact form

$$Q = q_0 \tau_z + \tau_x \mathbf{a} \cdot \mathbf{s} + \mathbf{b} \cdot \mathbf{s} + q_3 \tau_y s_z, \quad (365)$$

with

$$q_0 = d(u + v|\mathbf{h}|^2), \quad (366)$$

$$\mathbf{a} = u\mathbf{g} + v(\mathbf{g} \cdot \mathbf{h})\mathbf{h}, \quad (367)$$

$$\mathbf{b} = u\mathbf{h} + v[d^2\mathbf{h} + (\mathbf{g} \cdot \mathbf{h})\mathbf{g}], \quad (368)$$

$$q_3 = vd(\mathbf{g} \times \mathbf{h})_z. \quad (369)$$

For

$$S(\theta) = \cos \theta \tau_z s_z + \sin \theta \tau_y, \quad (370)$$

the auxiliary operator

$$\tilde{S} = -\frac{1}{2}\{Q, S\} \quad (371)$$

reduces to

$$\tilde{S} = -[A_\theta s_z + \tau_y \mathbf{C}_\theta \cdot \mathbf{s}], \quad (372)$$

where

$$A_\theta = q_0 \cos \theta + q_3 \sin \theta, \quad (373)$$

and

$$\mathbf{C}_\theta = \cos \theta \begin{pmatrix} a_y \\ -a_x \end{pmatrix} + \sin \theta \begin{pmatrix} b_x \\ b_y \end{pmatrix}. \quad (374)$$

Since s_z anticommutes with the in-plane spin matrices, Eq. (372) immediately gives

$$\boxed{\tilde{S}^2 = \delta^2(\mathbf{k}, \theta) \mathbb{1}}, \quad (375)$$

with

$$\delta^2 = A_\theta^2 + |\mathbf{C}_\theta|^2. \quad (376)$$

Thus the auxiliary spectrum remains doubly degenerate,

$$\text{spec } \tilde{S} = \{-\delta, -\delta, +\delta, +\delta\}. \quad (377)$$

Using $Q^2 = 1$, the angular dependence can be reduced further to

$$\delta^2(\mathbf{k}, \theta) = \frac{1}{2} [1 + \cos 2\vartheta(\mathbf{k}) \cos 2\theta + \sin 2\vartheta(\mathbf{k}) \sin 2\theta], \quad (378)$$

where the preferred local grading angle is determined directly by the Hamiltonian,

$$\boxed{\cos 2\vartheta = \frac{d^2 + |\mathbf{g}|^2 - |\mathbf{h}|^2}{E_+ E_-}}, \quad \boxed{\sin 2\vartheta = \frac{-2(\mathbf{g} \times \mathbf{h})_z}{E_+ E_-}}. \quad (379)$$

These expressions are properly normalized because

$$\boxed{(E_+ E_-)^2 = (d^2 + |\mathbf{g}|^2 - |\mathbf{h}|^2)^2 + 4(\mathbf{g} \times \mathbf{h})_z^2}. \quad (380)$$

Consequently,

$$\boxed{\delta(\mathbf{k}, \theta) = |\cos[\theta - \vartheta(\mathbf{k})]|}. \quad (381)$$

Equation (381) is the main simplification. The simple Rashba operator K_R is no longer conserved, but the auxiliary problem still reduces to a single preferred angle $\vartheta(\mathbf{k})$ at each momentum.

c. Possible local grading operators. Before optimizing the grading, it is useful to ask how general the onsite operator S should be. The local Hilbert space contains two orbital and two spin degrees of freedom, so any Hermitian onsite operator can be expanded in the sixteen matrices

$$\tau_\mu s_\nu, \quad \mu, \nu = 0, x, y, z. \quad (382)$$

The grading is required to be odd under time reversal,

$$TST^{-1} = -S, \quad T = 1s_y K. \quad (383)$$

Among the sixteen Hermitian matrices, the ten time-reversal-odd directions are

$$\boxed{s_i, \quad \tau_x s_i, \quad \tau_z s_i, \quad \tau_y, \quad i = x, y, z.} \quad (384)$$

Thus, before imposing any further constraint, a general local time-reversal-odd grading could involve ten real coefficients.

The condition $S^2 = 1$ strongly reduces this freedom. Rather than parameterizing the full nonlinear space of such matrices, consider small deformations of the one-angle family

$$S(\theta) = \cos \theta \tau_z s_z + \sin \theta \tau_y. \quad (385)$$

For

$$S \rightarrow S + \delta S, \quad (386)$$

the condition $S^2 = 1$ requires, to linear order,

$$\{S, \delta S\} = 0. \quad (387)$$

Within the ten time-reversal-odd matrices in Eq. (384), this leaves six independent tangent directions. A convenient basis is

$$\begin{aligned} G_1 &= \sin \theta \tau_x s_x + \cos \theta s_y, \\ G_2 &= \sin \theta \tau_x s_y - \cos \theta s_x, \\ G_3 &= \tau_x s_z, \\ G_4 &= \tau_z s_x, \\ G_5 &= \tau_z s_y, \\ G_6 &= \cos \theta \tau_y - \sin \theta \tau_z s_z. \end{aligned} \quad (388)$$

The last direction is simply

$$G_6 = \partial_\theta S(\theta), \quad (389)$$

and therefore corresponds to changing the angle already present in $S(\theta)$.

The Rashba–Dresselhaus Hamiltonian also preserves a twofold rotation about the z axis, C_{2z} . Under this symmetry, the tangent directions G_1, G_2, G_4, G_5 are odd, whereas G_3 and G_6 are even. If the grading is required to respect the same symmetry, the only genuinely new local direction, besides changing θ , is therefore

$$\boxed{\Gamma = G_3 = \tau_x s_z.} \quad (390)$$

This motivates the normalized two-angle extension

$$\boxed{S(\theta, \phi) = \cos \phi (\cos \theta \tau_z s_z + \sin \theta \tau_y) + \sin \phi \tau_x s_z.} \quad (391)$$

The three matrices $\tau_z s_z$, τ_y , and $\tau_x s_z$ square to unity and mutually anticommute, so $S(\theta, \phi)^2 = 1$ automatically.

The extra direction turns out not to improve the auxiliary gap. For the general in-plane Hamiltonian Eq. (350),

$$\{H, \Gamma\} = 0. \quad (392)$$

Spectral flattening preserves this anticommutation relation, so

$$\{Q, \Gamma\} = 0. \quad (393)$$

Consequently,

$$\begin{aligned}\tilde{S}(\theta, \phi) &= -\frac{1}{2}\{Q, S(\theta, \phi)\} \\ &= \cos \phi \tilde{S}(\theta).\end{aligned}\tag{394}$$

The corresponding auxiliary gap is therefore

$$\Delta_{\tilde{S}}(\theta, \phi) = |\cos \phi| \Delta_{\tilde{S}}(\theta).\tag{395}$$

It is maximal at $\phi = 0$. Thus, although the symmetry analysis allows one additional onsite grading direction, it cannot improve the auxiliary gap, and the optimal grading remains within the one-angle family $S(\theta)$.

d. Range of preferred grading angles. It remains to determine the range of $\vartheta(\mathbf{k})$ across the Brillouin zone. For $r^2 = p_x^2 + p_y^2 \neq 0$, introduce

$$s = \frac{2p_x p_y}{p_x^2 + p_y^2}, \quad -1 \leq s \leq 1,\tag{396}$$

and

$$t = \frac{d^2}{4(p_x^2 + p_y^2)} \geq 0.\tag{397}$$

For the Rashba–Dresselhaus textures in Eq. (351),

$$|\mathbf{g}|^2 = 4A^2 r^2,\tag{398}$$

$$|\mathbf{h}|^2 = 4r^2 (\lambda_R^2 + \lambda_D^2 + 2\lambda_R \lambda_D s),\tag{399}$$

$$(\mathbf{g} \times \mathbf{h})_z = -4Ar^2 (\lambda_R + \lambda_D s).\tag{400}$$

Since the denominator in Eq. (379) is positive, 2ϑ is the argument of

$$Z(s, t) = X(s, t) + iY(s),\tag{401}$$

with

$$X(s, t) = t + A^2 - \lambda_R^2 - \lambda_D^2 - 2\lambda_R \lambda_D s,\tag{402}$$

$$Y(s) = 2A(\lambda_R + \lambda_D s).\tag{403}$$

At the time-reversal-invariant momenta, where $r = 0$, the spin-dependent terms vanish and $\vartheta = 0$ by continuity.

For $A, \lambda_R, \lambda_D > 0$, consider first

$$\vartheta_+ = \arctan \frac{\lambda_R + \lambda_D}{A}.\tag{404}$$

Rotating Z by this angle gives

$$\begin{aligned}\text{Im}(Ze^{-2i\vartheta_+}) &= \frac{2A}{A^2 + (\lambda_R + \lambda_D)^2} \\ &\times [\lambda_D(s - 1)(A^2 + \lambda_R^2 - \lambda_D^2) - (\lambda_R + \lambda_D)t].\end{aligned}\tag{405}$$

In the gapped regime

$$\boxed{\lambda_D^2 < A^2 + \lambda_R^2},\tag{406}$$

the right-hand side of Eq. (405) is non-positive because $s \leq 1$ and $t \geq 0$. Hence

$$\vartheta(\mathbf{k}) \leq \vartheta_+ = \arctan \frac{\lambda_R + \lambda_D}{A}.\tag{407}$$

The bound is saturated at

$$s = 1, \quad t = 0,\tag{408}$$

corresponding to $p_x = p_y$ and $d(\mathbf{k}) = 0$.

The lower bound depends on the relative strengths of Rashba and Dresselhaus coupling. If

$$0 \leq \lambda_D \leq \lambda_R, \quad (409)$$

then

$$Y(s) = 2A(\lambda_R + \lambda_D s) \geq 0 \quad (410)$$

for every $s \in [-1, 1]$. On the branch continuously connected to the pure Rashba model this gives

$$\vartheta(\mathbf{k}) \geq 0. \quad (411)$$

Since $\vartheta = 0$ at the time-reversal-invariant momenta,

$$\vartheta_{\min} = 0. \quad (412)$$

When $\lambda_D > \lambda_R$, define instead

$$\vartheta_- = \arctan \frac{\lambda_R - \lambda_D}{A} < 0. \quad (413)$$

A second rotation gives

$$\begin{aligned} \text{Im}(Ze^{-2i\vartheta_-}) &= \frac{2A}{A^2 + (\lambda_R - \lambda_D)^2} \\ &\times [\lambda_D(s+1)(A^2 + \lambda_R^2 - \lambda_D^2) + (\lambda_D - \lambda_R)t]. \end{aligned} \quad (414)$$

Under the same condition Eq. (406), this expression is non-negative. Therefore

$$\vartheta(\mathbf{k}) \geq \vartheta_- = \arctan \frac{\lambda_R - \lambda_D}{A}. \quad (415)$$

The lower bound is saturated at $s = -1$ and $t = 0$, corresponding to $p_y = -p_x$ and $d(\mathbf{k}) = 0$.

Provided the contour $d(\mathbf{k}) = 0$ intersects the two diagonals, the bounds are therefore attained and

$$\boxed{\vartheta_{\max} = \arctan \frac{\lambda_R + \lambda_D}{A}}, \quad (416)$$

while

$$\boxed{\vartheta_{\min} = \begin{cases} 0, & 0 \leq \lambda_D \leq \lambda_R, \\ \arctan \frac{\lambda_R - \lambda_D}{A}, & \lambda_R < \lambda_D < \sqrt{A^2 + \lambda_R^2}. \end{cases}} \quad (417)$$

For the square-lattice mass $d(\mathbf{k}) = M + 2A(\cos k_x + \cos k_y)$, the $d = 0$ contour intersects both diagonals whenever $|M| < 4A$. For the parameters used below, $M = -2A$, the intersections occur at $k_x = \pm k_y = \pm\pi/3$.

The condition Eq. (406) also has a direct physical meaning. At $\lambda_D^2 = A^2 + \lambda_R^2$, the two in-plane spin textures can become parallel with equal magnitude on the $d = 0$ contour, allowing the lower physical band gap E_- to close. The bounds above thus apply precisely on the branch connected continuously to the gapped Rashba model.

e. Optimal fixed grading. Equation (381) reduces the global optimization to a simple maximin problem. A single onsite grading must approximate all preferred angles in the interval $[\vartheta_{\min}, \vartheta_{\max}]$. The optimal choice is the centre of this interval,

$$\boxed{\theta_{\text{opt}} = \frac{\vartheta_{\min} + \vartheta_{\max}}{2}}, \quad (418)$$

and the corresponding auxiliary gap is

$$\boxed{\Delta_{\tilde{S}}^{\text{opt}} = \cos \left[\frac{\vartheta_{\max} - \vartheta_{\min}}{2} \right]}. \quad (419)$$

For $0 \leq \lambda_D \leq \lambda_R$, this gives

$$\theta_{\text{opt}} = \frac{1}{2} \arctan \frac{\lambda_R + \lambda_D}{A}. \quad (420)$$

For $\lambda_R < \lambda_D < \sqrt{A^2 + \lambda_R^2}$,

$$\theta_{\text{opt}} = \frac{1}{2} \left[\arctan \frac{\lambda_R + \lambda_D}{A} + \arctan \frac{\lambda_R - \lambda_D}{A} \right]. \quad (421)$$

Finally,

$$S_{\text{opt}} = \cos \theta_{\text{opt}} \tau_z s_z + \sin \theta_{\text{opt}} \tau_y. \quad (422)$$

The pure Rashba result is recovered at $\lambda_D = 0$. Dresselhaus coupling therefore destroys the simple Rashba sector operator, but it does not destroy the one-angle grading construction. Instead, it broadens and, for $\lambda_D > \lambda_R$, shifts the range of preferred local grading angles across the Brillouin zone. The optimal onsite grading remains the centre of that range.

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